fifo_memory

fifo_mem

CLK

clk_wr_IBUF_BUFG

data_out0

rst_rd

rst_rd_IBUF wr_ptr0

data_in_top_IBUF[7:0]

data_out[7:0]

data_out_reg[1]_0[2:0]

out[2:0]

mem_reg_0_7_0_5

RAM32M

WCLK WE

DOA[1:0]

DOB[1:0]

DOC[1:0]

DOD[1:0] n/c

ADDRA[4:0]

ADDRB[4:0]

ADDRC[4:0] ADDRD[4:0] DIA[1:0] DIB[1:0]

DIC[1:0]

DID[1:0]

mem_reg_0_7_6_7

RAM32X1D

DPO
SPO n/c

A0A1A2A3A4D

DPRA0 DPRA1 DPRA2 DPRA3 DPRA4 WCLK WE

mem_reg_0_7_6_7__0

RAM32X1D

DPO
SPO n/c

A0A1A2A3A4D

DPRA0 DPRA1 DPRA2 DPRA3 DPRA4 WCLK WE

read_pointer

read_ptr

CLK

data_out0
empty_OBUF

rd_gptr_reg[0]_0

rd_gptr_reg[3]_0

read_en_IBUF

Q[2:0]

out[2:0]

rd_gptr_reg[3]_1[3:0]

sync_read

FF_sync

CLK

full_OBUF

p_1_in1_in

rst_wr

rst_wr_IBUF

wr_ptr0

write_en_IBUF

D[3:0]

Q[2:0]

sync_write

FF_sync_0

CLK

p_1_in1_in
sync_ff2_reg[3]_1

Q[2:0]

sync_ff2_reg[3]_0[3:0] write_pointer

write_ptr

CLK

p_1_in1_in wr_gptr_reg[3]_0 wr_ptr0

Q[2:0]

out[2:0]

data_in_top_IBUF[1]_inst

IBUF I O
data_in_top_IBUF[2]_inst

IBUF I O
data_in_top_IBUF[3]_inst

IBUF I O
data_in_top_IBUF[4]_inst

IBUF I O
data_in_top_IBUF[5]_inst

IBUF I O
data_in_top_IBUF[6]_inst

IBUF I O
data_in_top_IBUF[7]_inst

IBUF I O

data_out_top[7:0] data_out_top_OBUF[0]_inst OBUF I O
data_out_top_OBUF[1]_inst OBUF I O
data_out_top_OBUF[2]_inst OBUF I O
data_out_top_OBUF[3]_inst OBUF I O
data_out_top_OBUF[4]_inst OBUF I O
data_out_top_OBUF[5]_inst OBUF I O
data_out_top_OBUF[6]_inst OBUF I O
data_out_top_OBUF[7]_inst OBUF I O
empty empty_OBUF_inst OBUF I O

wr_ptr_reg[2]

FDCE Q C

CECLR D

data_out_reg[0]

FDCE Q C

CECLR D

data_out_reg[1]

FDCE Q C

CECLR D

data_out_reg[2]

FDCE Q C

CECLR D

data_out_reg[3]

FDCE Q C

CECLR D

data_out_reg[4]

FDCE Q C

CECLR D

data_out_reg[5]

FDCE Q C

CECLR D

data_out_reg[6]

FDCE Q C

CECLR D

data_out_reg[7]

FDCE Q C

CECLR D

wr_ptr_reg[1]

FDCE Q C

CECLR D

RAMA

RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WERAMA_D1
RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WE RAMB
RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WERAMB_D1
RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WE RAMC
RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WERAMC_D1
RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WE RAMD
RAMS32 O

ADR0 ADR1 ADR2 ADR3 ADR4 CLK IWERAMD_D1
RAMS32 O

ADR0 ADR1 ADR2 ADR3 ADR4 CLK IWE

wr_ptr_reg[0]

FDCE Q C

CECLR D

wr_ptr[2]_i_1

LUT3 O

I0
I1
I2

DP

RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WE

SP

RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WE

DP

RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WE

SP

RAMD32 O

CLK IRADR0 RADR1 RADR2 RADR3 RADR4 WADR0 WADR1 WADR2 WADR3 WADR4 WE

rd_gptr[2]_i_3
LUT1

I0 O

full full_OBUF_inst OBUF I O

read_en

read_en_IBUF_inst

IBUF I O

wr_ptr[1]_i_1
LUT2

I0 O
I1

empty_OBUF_inst_i_1

LUT3 O

I0
I1

I2 empty_OBUF_inst_i_2

LUT6 O

I0
I1
I2
I3
I4
I5

rd_gptr[0]_i_1
LUT1

I0 O

rd_gptr[1]_i_1

LUT3 O

I0
I1
I2

rd_gptr[2]_i_1

LUT4 O

I0
I1
I2
I3

rd_gptr[2]_i_2

LUT4 O

I0
I1
I2
I3

rd_gptr[3]_i_1

LUT4 O

I0
I1
I2
I3

rd_gptr_reg[0]

FDCE Q C

CECLR D

rd_gptr_reg[1]

FDCE Q C

CECLR D

rd_gptr_reg[2]

FDCE Q C

CECLR D

rd_gptr_reg[3]

FDCE Q C

CECLR D

rd_ptr[0]_i_1
LUT1

I0 O

rd_ptr[1]_i_1

LUT2

I0 O
I1

rd_ptr[2]_i_1

LUT3 O

I0
I1
I2

rd_ptr_reg[0]

FDCE Q C

CECLR D

rd_ptr_reg[1]

FDCE Q C

CECLR D

rd_ptr_reg[2]

FDCE Q C

CECLR D

rst_rd

rst_rd_IBUF_inst

IBUF I O

rst_wr

rst_wr_IBUF_inst

IBUF I O

wr_ptr[0]_i_1
LUT1

I0 O

full_OBUF_inst_i_1

LUT3 O

I0
I1
I2

full_OBUF_inst_i_2

LUT6 O

I0
I1
I2
I3
I4
I5

sync_ff1[3]_i_1
LUT1

I0 O

sync_ff1_reg[0]

FDCE Q C

CECLR D

sync_ff1_reg[1]

FDCE Q C

CECLR D

sync_ff1_reg[2]

FDCE Q C

CECLR D

sync_ff1_reg[3]

FDCE Q C

CECLR D

sync_ff2_reg[0]

FDCE Q C

CECLR D

sync_ff2_reg[1]

FDCE Q C

CECLR D

sync_ff2_reg[2]

FDCE Q C

CECLR D
sync_ff2_reg[3]

FDCE Q C

CECLR D

wr_gptr[2]_i_1

LUT4 O

I0
I1
I2
I3

wr_gptr_reg[3]

FDCE Q C

CECLR D

sync_ff1_reg[0]

FDCE Q C

CECLR D

sync_ff1_reg[1]

FDCE Q C

CECLR D

sync_ff1_reg[2]

FDCE Q C

CECLR D

sync_ff1_reg[3]

FDCE Q C

CECLR D

sync_ff2_reg[0]

FDCE Q C

CECLR D

sync_ff2_reg[1]

FDCE Q C

CECLR D

sync_ff2_reg[2]

FDCE Q C

CECLR D

sync_ff2_reg[3]

FDCE Q C

CECLR D

write_en write_en_IBUF_inst

IBUF I O

wr_gptr_reg[2]

FDCE Q C

CECLR D

wr_gptr[0]_i_1
LUT1

I0 O

wr_gptr[1]_i_1

LUT3 O

I0
I1
I2

wr_gptr[2]_i_2

LUT4 O

I0
I1
I2
I3

wr_gptr[3]_i_1

LUT4 O

I0
I1
I2
I3

wr_gptr_reg[0]

FDCE Q C

CECLR Dwr_gptr_reg[1]
FDCE Q C

CECLR D

data_in_top_IBUF[0]_inst

IBUF I O

data_in_top[7:0]

clk_wr_IBUF_inst

IBUF I O

clk_wr_IBUF_BUFG_inst BUFG I O

clk_wr

clk_rd_IBUF_inst

IBUF I O

clk_rd_IBUF_BUFG_inst BUFG I O

clk_rd